

# Selecting an early-warning model on the OULAD graph

*A step-by-step record of the process, the measurements, and the corrections*

OULAD graph analytics — repository `jose-alvarado-guzman/oulad`  
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## 0. What this document is

A walkthrough of the model-selection process in the order it was actually carried out: what was built at each step, what it measured, what the numbers were, and which conclusions had to be withdrawn afterwards. Every figure came from a run against a live AuraDB instance and an Aura Graph Analytics session, or from an offline script over the source CSVs. Nothing is estimated.

`docs/model-selection.md` remains the source of record for the figures; this document is the narrative walkthrough of the same work. Where they disagree, the shorter one in the repository is newer.

**Excluded by request:** the weighted node-similarity and Louvain cohort analysis in `notebooks/aga_student_cohorts.ipynb`. That work is descriptive rather than predictive — it produces no held-out metric — so nothing below depends on it. One finding of it is carried forward, because a later step rests on it: the discovery that a large group of registered students never touch a material at all and therefore cannot be embedded.

The process ran in eight steps:

| step | what it established                                             |
|------|-----------------------------------------------------------------|
| 1    | the question, the target, and what would count as leakage       |
| 2    | the baselines any model has to beat, and the metric to judge on |
| 3    | FastRP topology embedding — rejected, +0.4 accuracy points      |
| 4    | FastPath sequence embedding, and <i>when</i> the signal arrives |
| 5    | replication on a second module                                  |
| 6    | the zero-activity rule — the cheap complement                   |
| 7    | assessment submission — the largest single gain measured        |
| 8    | persist the model, score a module it has never seen             |

## 1. The question, the target, and the leakage boundary

**Question.** Given the OULAD graph, can a student's final result be predicted from their position in it — which materials they engaged with, in what order, and which demographic groups they belong to — early enough for the prediction to be actionable?

**Target.** Binary: `passed = 1` for `Pass` or `Distinction`, `0` for `Fail` or `Withdrawn`, taken from `finalResult` on `(:StudentRegistration)-[:CONTAINS_COURSE]->(:Course)`.

**Primary module.** GGG throughout, except where noted — 2,531 registered students, 2,359 of whom touched at least one material, across 281,277 interactions.

## 1.1 What was excluded, and why

**Assessment scores.** `WAS_ASSESSED_IN` carries marks, and marks determine `finalResult` almost by definition. Predicting the outcome from the marks that produced it is leakage in a costume. Every model below sees engagement and demographics only.

Two further quantities sit closer to that line than they look, and both are called out in the notebooks:

- `dateUnregistration` on `CONTAINS_COURSE` is effectively the withdrawal label.
- Interactions recorded *after* a student stopped participating leak the same information backwards. This turned out to matter enormously — it is what step 4 exists to measure.

**The boundary that was later reconsidered.** Excluding scores also discarded *whether a student submitted at all*, which is behaviour rather than outcome. Step 7 revisits that and is where the largest measured gain came from. The test applied there: not "is it correlated with the outcome" — clicks are too — but "is it observable at the decision point without the label".

## 1.2 A structural blind spot, and a bad argument for it

**1,287 of 7,909 BBB registrations never touched a material**, and 88.0% of them withdrew against 19.0% of engaged students. A student with no interactions has no edges, so cannot be projected, embedded or classified. **Every result in this document is conditional on having engaged at all**, and the excluded group is the one an intervention would most want to reach.

That 88.0% is a real number and a bad argument, which took a dedicated measurement to establish: **790 of those 1,287 students (61.4%) unregistered before day 0**. It largely says "students who quit before teaching began were recorded as withdrawn." It says nothing about what is visible at a decision point, and it had been cited in this repository as support for a day-14 rule that nobody had measured. Step 6 measures the rule properly.

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## 2. The bar, and choosing a measure

### 2.1 Baselines

Any model has to beat these. Both computed over the same population the models see.

| baseline                                              | F1 macro | accuracy |
|-------------------------------------------------------|----------|----------|
| always predict the majority class                     | —        | 0.6393   |
| <code>clicks &gt;= median</code> , no model at all    | 0.7635   | 0.7681   |
| <code>logClicks</code> through the same random forest | 0.8436   | 0.8658   |

The third row is the one that matters. **A single logged aggregate, given a decent model, reaches 0.8658** — and it is the reason several apparently good results below are not good at all.

## 2.2 The metric, which was got wrong twice

The same FastPath embedding was judged three ways and got three verdicts. Louvain modularity said it had failed; accuracy said it was mediocre early; precision said it produces a usable worklist. None of the measurements were wrong — two of them were answering a question nobody was asking.

What an intervention team needs is not accuracy. Accuracy counts every unflagged failure against the model, which punishes a conservative classifier for being conservative, and it is dominated by the majority class on an imbalanced target. The operational quantities are:

- **flagged** — how many students land on the worklist.
- **recall** — what share of eventual failures that catches.
- **precision** — what share of the flags are real.
- **lift** — precision divided by the base rate of the population it was measured on,  $(\text{hit} / k) / \text{base\_rate}$ . A lift of 1.0 means the ranking adds nothing over picking the same number of students at random; below 1.0 it is worse than random. The column exists because **precision alone does not travel between populations**: 0.93 is strong where 27% of students fail and below chance where 90% do.
- **precision at k** — precision restricted to the top  $k$  students by predicted risk, where  $k$  is the contact budget. Lift falls and recall rises as  $k$  grows, so a result is quoted as a budget, never as a single number.

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## 3. Step 3 — FastRP topology embedding

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`notebooks/aga_outcome_prediction.ipynb`.

### 3.1 What was built

FastRP over each student's engagement and demographic neighbourhood — `REVIEWED_MATERIAL`, `IN_AGE_GROUP`, `LIVE_IN_REGION`, `HAS_HIGHER_EDUCATION`, `IN_DEPRIVATION_GROUP` — fed to a GDS node classification pipeline with two candidates (logistic regression, random forest) selected on F1 macro over four validation folds. Held-out 30% split, 2,525 students.

The class label is computed **inside the projection query** and attached as a node property, so nothing is written to the database; it exists only in the session. `WAS_ASSESSED_IN` is absent by the rule in step 1.

`REVIEWED_MATERIAL` is left unaggregated, so a student who returned to a page many times has many parallel edges to it — implicit frequency weighting for the random projection.

### 3.2 The ablation, which is the whole point

Three arms, because a single number cannot attribute a gain:

| features                | F1 macro | accuracy |
|-------------------------|----------|----------|
| embedding only          | 0.3720   | 0.5923   |
| volume only (logClicks) | 0.8428   | 0.8549   |
| embedding + volume      | 0.8475   | 0.8588   |

**The embedding alone is a constant classifier.** It predicted "pass" for all 2,525 students — 1,508 right, 1,017 wrong. F1 macro 0.372 is exactly what a majority-class guess scores.

**The mechanism.** FastRP normalises its embeddings, which suppresses degree, and degree is the signal here. An embedding built from this graph knows *which* materials a student touched but barely encodes *how many*. Handing the model one logged click total fixes it completely, at which point it catches 751 of the 1,017 at-risk students. Volume is logged because raw click totals run from 0 to 10,768 and that skew is poison for logistic regression.

### 3.3 Verdict

**Rejected.** The embedding contributes **+0.4 accuracy points** over one logged feature — noise on a 760-student split. The working model is real and useful; FastRP is not what makes it work.

The ablation is kept in the notebook deliberately. Without it, the notebook reads as though 128 dimensions bought the +5.7 points over the median-threshold baseline, when a random forest on one logged feature did.

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## 4. Step 4 — FastPath sequence embedding, and when the signal arrives

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`notebooks/aga_fastpath_journeys.ipynb`.

### 4.1 Building a sequence the graph does not have

The loaded graph has no order in it, so the notebook builds one:

```
(:Student)-[:FIRST_INTERACTION]->(:Interaction)-[:NEXT_INTERACTION]->...
```

one `Interaction` node per (student, material, day) in date order — 281,277 nodes for GGG, deleted afterwards. FastPath embeds each chain; the embedding is then scored as a classifier feature.

Four things about that build are load-bearing:

**Numeric event features must be passed explicitly.** They go in through `event_node_feature_vector_property` as a **list**, even for a single number. Leaving it unset means click intensity never reaches the algorithm — a page opened once and one hammered fifty times become identical events.

**The activity-type vocabulary must be enumerated globally, not per module.** GGG has 7 activity types, BBB 12, EEE 11, out of 20 in the dataset. `activityTypeId` is a *categorical* input, so a per-module enumeration makes 0 mean `forumng` in GGG and `dualpane` in EEE; the embedding spaces are then unrelated and a model trained on one module scores another confidently and meaninglessly, with no error anywhere. This becomes critical in step 8.

**A node or list element pulled out of a map cannot be used directly inside a `CREATE` pattern.** `CREATE (ev)-[:OF_MATERIAL]->(event.material)` and `CREATE (s)-[:FIRST]->(chain[0])` are both syntax errors; bind with a `WITH` first. `EXPLAIN` catches this without writing anything.

`sessions.estimate()` under-sizes `FastPath`. For a 113k-node / 220k-relationship chain it returned `m_2GB` and `FastPath` aborted mid-run with *"The job ran out of memory"*. The cascade is worse than the cause: no embedding is written, so the next cell fails with *"Node properties [journeyEmbedding] do not exist in the graph"*, which reads as a pipeline bug rather than a memory one. The notebook now overrides the estimate with a floor by node count (8/16/32 GB).

## 4.2 The cutoff sweep — the decisive experiment

`CUTOFF_DAY` limits how much of each journey the model may see, so the sweep answers *when* the information arrives rather than only whether a model works.

Accuracy as GDS reports it, GGG, on its own internal 30% test split:

| cutoff        | students | majority | threshold | volume | journey       | both   |
|---------------|----------|----------|-----------|--------|---------------|--------|
| day 30        | 2,199    | 0.6462   | 0.5703    | 0.6439 | <b>0.6348</b> | 0.6455 |
| day 60        | 2,314    | 0.6422   | 0.5808    | 0.6029 | <b>0.6892</b> | 0.6978 |
| day 90        | 2,342    | 0.6405   | 0.6161    | 0.6515 | <b>0.7127</b> | 0.7226 |
| whole journey | 2,359    | 0.6393   | 0.7681    | 0.8658 | <b>0.9280</b> | 0.9308 |

F1 macro for the journey embedding: 0.5198 → 0.5980 → 0.6449 → 0.9176.

**Sweep methodology.** The sweep projects **day prefixes** of one chain rather than rebuilding four times. Because the chain is ordered by day, a prefix by day is exactly the graph a truncated build produces — a `tgt.day <= cutoff + SHIFT` filter is the whole trick. Verified: the day-30 and whole-journey points reproduced their independent rebuilds *exactly* (0.6348 and 0.9280) on byte-identical projections of 64,762 and 283,636 nodes. `logClicks`, the threshold baseline and the majority floor are recomputed per window, so no cutoff is scored against another's hindsight.

**When changing `CUTOFF_DAY`, the cutoff must reach every feature.** It gates the chain build, `LABEL_QUERY`'s `logClicks` and `BASELINE_QUERY`. Missing any one pits a truncated feature against a hindsight one — caught before running, but only just.

## 4.3 The table that should drive the decision

Evaluated on a **707-student holdout the model never trained on** (30%, seed 42), journey + volume:

| cutoff        | at risk | flagged | recall       | precision    | accuracy |
|---------------|---------|---------|--------------|--------------|----------|
| day 30        | 234     | 112     | 0.231        | 0.482        | 0.635    |
| <b>day 90</b> | 258     | 133     | <b>0.372</b> | <b>0.722</b> | 0.717    |
| whole journey | 261     | 204     | 0.759        | 0.971        | 0.902    |

Volume alone on the same holdout reaches precision 0.382, 0.417 and 0.825. **The embedding nearly doubles it at day 90**, which is the clearest evidence in this exercise that the sequence carries something an aggregate does not — and it survives an honest split.

## 4.4 The split had to be built by hand

GDS splits internally but does not expose which nodes its test set held, so `predict_stream` over all students mixes training data into the evaluation. The split here is explicit: students carry an `isHoldout` flag projected as a node property, the pipeline trains on a `TrainStudent` label and predicts on `HoldoutStudent`.

**Order matters.** `fast_path.mutate` registers its output property against `base_node_label` only, so a `TrainStudent` label that existed at projection time does not carry `journeyEmbedding`, and training dies with "node properties do not exist in the graph or part of the pipeline". The labels must be added with `gds.graph.node_labels.mutate` **after** the embedding step.

## 4.5 How much the in-sample figures had overstated it

| cutoff        | precision gap | recall gap | accuracy gap |
|---------------|---------------|------------|--------------|
| day 30        | <b>+0.450</b> | +0.356     | +0.206       |
| day 90        | +0.165        | +0.144     | +0.090       |
| whole journey | +0.017        | +0.068     | +0.033       |

Day 30 was almost entirely memorisation: 0.932 precision in-sample against 0.482 held out.

Note the shape of those gaps. **Accuracy moved 3–21 points while precision moved up to 45.** Accuracy is dominated by the majority class and conceals memorisation, so "the accuracy gap is small, therefore the inflation is small" is not a valid argument — and it was the argument that let a bad day-30 recommendation stand for one revision of the record. The reassurance was circular on top of that: the accuracy column in the sweep is GDS's own held-out metric, so it *could not* show inflation.

## 4.6 Three readings

**On GGG the signal starts between day 30 and day 60.** At day 30 the embedding sits *below* the majority floor — nothing works. By day 60 it is 4.7 points above it and still climbing.

**Inside that window, sequence beats volume decisively.** Journey minus volume, in accuracy points: day 30 –0.9, **day 60 +8.6**, day 90 +6.1, whole journey +6.2. At day 60 click volume alone (0.6029) is *worse than guessing*, while the shape of the same two months reaches 0.6892.

**The untruncated score is mostly hindsight.** 0.7127 at day 90 against 0.9280 at the end: about two thirds of the apparent skill arrives after month three of a nine-month presentation, which is largely the model noticing who stopped. For a withdrawal, "when did activity stop" is the label restated. So the untruncated model is a strong retrospective classifier and a poor early-warning one, and the two claims are not interchangeable.

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## 5. Step 5 — replication on a second module

Everything above is GGG. Repeating the identical procedure on **BBB** — 6,484 students, 1,139,085 events, days –23..268, same seed, same 30% holdout — gives 1,938 held-out students at day 90:

| cutoff        | features    | flagged    | of at risk | recall       | precision    | accuracy | precision gap |
|---------------|-------------|------------|------------|--------------|--------------|----------|---------------|
| day 30        | volume      | 549        | 812        | 0.425        | 0.628        | 0.650    | −0.015        |
| day 30        | both        | 446        | 812        | 0.386        | 0.702        | 0.670    | +0.141        |
| day 90        | volume      | 591        | 826        | 0.504        | 0.704        | 0.698    | −0.010        |
| <b>day 90</b> | <b>both</b> | <b>520</b> | <b>826</b> | <b>0.528</b> | <b>0.838</b> | 0.755    | <b>+0.015</b> |
| whole journey | volume      | 735        | 829        | 0.679        | 0.766        | 0.775    | −0.011        |
| whole journey | both        | 677        | 829        | 0.776        | 0.950        | 0.887    | +0.023        |

Four findings:

**The day-90 recommendation replicates, and more cleanly than on GGG.** Precision 0.838 against GGG's 0.722, recall 0.528 against 0.372, inflation gap +0.015 against +0.165.

**At day 90 the embedding dominates volume outright in both modules** — not a precision/recall trade. On BBB it flags *fewer* students than volume (520 against 591) and catches *more* failures (436 against 416). On GGG, 133 flags against 163 and 96 caught against 68. That is the single most robust result in this exercise.

**GGG's day-30 collapse does not generalise.** On BBB, day 30 holds out at 0.702 precision with a +0.141 gap. Day 30 on GGG was memorisation; day 30 on BBB is signal. "The embedding needs 60 days" is a statement about GGG, not about the method.

**Only the embedding inflates, and mechanically so.** Across all six BBB configurations the volume-only gaps are −0.015 to −0.010 — negative, i.e. the holdout scored *better* than the training rows, which is what noise around zero inflation looks like. Every positive gap belongs to a model carrying `journeyEmbedding`. One logged feature cannot memorise 4,500 training rows; 129 features can. Any figure quoted for an embedding model without a holdout should be assumed inflated; the volume-only baselines never needed the correction.

**But the absolute numbers are not transferable.** Day-90 precision is 0.722 on GGG and 0.838 on BBB for identical code and hyperparameters. Quote a range or re-measure per module; do not carry one module's figure across.

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## 6. Step 6 — the zero-activity rule, measured properly

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`scripts/zero_activity_rule.py`, offline, no database. Full record in `docs/early-warning-rule.md`.

A student still registered who has not touched a single material by day  $D$  is at risk. One `NOT EXISTS` clause, no session, no embedding, weeks before any model here can speak. This step exists because the claim that motivated the rule (§1.2) did not support it.

### 6.1 It does not predict withdrawal

Among students still registered at the threshold, against `finalResult = 'Withdrawn'`:

| day | population | flagged | % of cohort | withdrawal rate | base rate | lift | recall |
|-----|------------|---------|-------------|-----------------|-----------|------|--------|
| 7   | 29,178     | 2,924   | 10.0%       | 0.340           | 0.219     | 1.55 | 0.147  |
| 14  | 28,119     | 1,691   | 6.0%        | 0.273           | 0.198     | 1.38 | 0.081  |
| 21  | 27,908     | 1,185   | 4.3%        | 0.277           | 0.193     | 1.44 | 0.060  |
| 28  | 27,538     | 921     | 3.3%        | 0.222           | 0.184     | 1.20 | 0.040  |

Lift 1.2 to 1.55 and recall under 15%. On its own that is a negative result and the end of the idea.

## 6.2 It predicts *failure*

Against `Withdrawn OR Fail` — the same target the day-90 model is trained on:

| day | flagged | % of cohort | precision    | base rate | lift | recall       | caught / at risk |
|-----|---------|-------------|--------------|-----------|------|--------------|------------------|
| 7   | 2,924   | 10.0%       | <b>0.736</b> | 0.473     | 1.66 | <b>0.156</b> | 2,151 / 13,793   |
| 14  | 1,691   | 6.0%        | 0.765        | 0.453     | 1.77 | 0.102        | 1,293 / 12,734   |
| 21  | 1,185   | 4.3%        | 0.813        | 0.449     | 1.88 | 0.077        | 963 / 12,523     |
| 28  | 921     | 3.3%        | 0.823        | 0.441     | 1.92 | 0.062        | 758 / 12,153     |

Roughly three of every four students still enrolled and silent go on to fail or withdraw, against a base rate of about 45%. **Day 7, not day 28:** precision rises only 8.7 points from day 7 to day 28 while recall falls by 60%, and the rule's entire value is lead time. Day 14 was the threshold originally proposed and is dominated by day 7 on both counts.

## 6.3 The overlap costs the model nothing

The worry was double-counting — that the model's precision comes largely from students the rule already flags for free. Measured on the same holdout, partitioned by whether the day-7 rule fired:

| module | population               | n     | day-90 model precision | recall |
|--------|--------------------------|-------|------------------------|--------|
| GGG    | whole holdout            | 702   | 0.722                  | 0.372  |
| GGG    | <b>rule did not flag</b> | 564   | <b>0.760</b>           | 0.398  |
| GGG    | rule already flagged     | 138   | 0.606                  | 0.299  |
| BBB    | whole holdout            | 1,938 | 0.838                  | 0.528  |
| BBB    | <b>rule did not flag</b> | 1,731 | <b>0.856</b>           | 0.511  |
| BBB    | rule already flagged     | 207   | 0.760                  | 0.635  |

**The model gets better, not worse, once the rule takes its share** — +3.8 points on GGG, +1.8 on BBB. The prediction going in was the reverse. The students the rule flags are the ones the model handles worst: silent for a week and then engaging, an irregular shape the embedding reads poorly. The two triggers are cleanly complementary.

Note also that most rule-flagged students are outside the model's reach entirely: the rule flags 567 GGG students, but only 138 appear in the holdout with a journey to embed. The rest never engage at all — the blind spot from §1.2, which is what this rule exists to cover.

## 6.4 The trap that makes this query silently wrong

22,521 of 32,593 `dateUnregistration` values are float `NaN`. None are `null`. pandas writes missing numerics as `NaN` and the driver stores them as real floats, so both of these are false for a missing value:

- `cc.dateUnregistration IS NULL`
- `cc.dateUnregistration > 14` — every comparison with `NaN` is false, including `<=`, `>=` and `=`

An `IS NULL OR > $day` test therefore excludes every student who never unregistered — 69% of the data, and precisely the students the rule is about. For GGG at day 14 that returns 49 flagged students instead of 391: an 8× undercount, no error, and a plausible-looking result. The guard is `isNaN(...)` alongside the null check, and it is needed by any query in this repository that null-checks a numeric column sourced from pandas.

Reconciliation against the offline measurement: GGG at day 14 gives 391 flagged / 2,444 population in Cypher against 395 / 2,449 in pandas. The four-student gap is students enrolled in two GGG presentations, which the graph collapses to one `Student` node and the CSV keeps as two rows.

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## 7. Step 7 — assessment submission, revisiting the leakage boundary

`scripts/assessment_submission*.py`. Full record in `docs/assessment-submission.md`.

Scores stay excluded. **Whether a student submitted at all** is behaviour, not outcome, and it is observable at the decision point: at day 33 you know who missed the TMA due on day 19, and you do not know `finalResult` until the presentation ends.

### 7.1 Phase 1 — is it additive to the day-7 rule?

The four cells at day 90, over the 25,562 students still registered:

| day-7 rule | missed all assessments | students     | share       | at-risk rate |
|------------|------------------------|--------------|-------------|--------------|
| no         | no                     | 22,012       | 86.1%       | 0.342        |
| no         | <b>yes</b>             | <b>1,312</b> | <b>5.1%</b> | <b>0.912</b> |
| yes        | yes                    | 823          | 3.2%        | 0.926        |
| yes        | no                     | 1,415        | 5.5%        | 0.497        |

**1,312 students at 91.2% at risk that the rule never sees.** The signal is additive.

The go criterion was fixed before measuring — the marginal cell must hold  $\geq 5\%$  of the cohort at  $\geq 0.75$  at-risk rate, in at least 4 of 7 modules — and it passed at exactly 4 of 7 (DDD, EEE, FFE, GGG). Worth being precise about how: the three failures are all on *share*, not rate, and **every module clears the rate bar**, from 0.773 to 0.984. Read it as "reliable but sometimes small".

An unplanned finding: students the day-7 rule flags who then **do** submit are only **49.7%** at risk against 92.6% for those who don't, so the rule alone over-flags that group. Combining the two gives a four-tier triage rather than two independent alarms.

## 7.2 The trigger in operation

Judged 14 days after the first assessment is due:

| module | fires at day | flagged | precision | base rate | lift | new vs rule | lead over day 90 |
|--------|--------------|---------|-----------|-----------|------|-------------|------------------|
| AAA    | 33           | 50      | 0.740     | 0.257     | 2.88 | 44          | 57d              |
| BBB    | 31           | 672     | 0.897     | 0.414     | 2.17 | 342         | 59d              |
| CCC    | 32           | 361     | 0.906     | 0.513     | 1.77 | 248         | 58d              |
| DDD    | 37           | 605     | 0.886     | 0.499     | 1.77 | 465         | 53d              |
| EEE    | 47           | 277     | 0.931     | 0.341     | 2.73 | 179         | 43d              |
| FFF    | 36           | 656     | 0.892     | 0.442     | 2.02 | 493         | 55d              |
| GGG    | 75           | 398     | 0.847     | 0.365     | 2.32 | 213         | 15d              |

It beats the day-90 model on precision in both modules the model has been measured on — 0.847 against 0.722 on GGG, 0.897 against 0.838 on BBB — and arrives 15 and 59 days earlier. Recall is lower (0.17–0.41): a sharper, smaller net.

**Why 14 days of grace.** Late submission is normal — 28.4% of all submissions arrive after the due date — and **three presentations record zero submissions on or before the due date** (CCC 2014B, CCC 2014J, DDD 2013B), with 99%+ arriving within a week. Judged on the due date the trigger flagged 100% of those cohorts with precision equal to the base rate: a result that looks like a finding and is an artefact.

## 7.3 Phase 2 — ported to Cypher and reconciled

**176 of 176 comparisons exact**, across 22 presentations × two evaluation days × four quantities (assessmentsDue, population, missedAll, missedAllAtRisk), against a gate of 1%.

Tighter than the zero-activity rule, which reconciled to within four students, and the reason is worth keeping: **scoping through Course on both codeModule and codePresentation sidesteps the dual-presentation collapse**. A Student is one node across presentations, but each StudentRegistration→Course edge is presentation-specific.

Which guards actually matter, measured by removing each one over four sample presentations:

| variant                               | population   | flagged    | precision    |
|---------------------------------------|--------------|------------|--------------|
| both guards (correct)                 | 6,335        | 765        | 0.851        |
| no isNaN on a.date                    | 6,335        | 765        | 0.851        |
| <b>no isNaN on dateUnregistration</b> | <b>1,155</b> | 217        | <b>1.000</b> |
| <b>no banked exclusion</b>            | 6,335        | <b>634</b> | 0.886        |

isNaN on dateUnregistration is critical — without it the population collapses by 82% and precision reads a perfect 1.000. The banked exclusion is real: 1,909 rows carry a score from a previous sitting. isNaN on a.date is *not* load-bearing, because a.date <= \$day already excludes NaN — the plan for this work claimed otherwise.

## 7.4 Phase 3 — as model features

Five arms, one harness, same 30% holdout and seed 42, day 90:

| module | features                      | precision    | recall | p@100       | p@200        |
|--------|-------------------------------|--------------|--------|-------------|--------------|
| BBB    | volume                        | 0.715        | 0.485  | 0.93        | 0.870        |
| BBB    | journey + volume              | 0.764        | 0.606  | 0.92        | 0.895        |
| BBB    | <b>submission</b>             | 0.831        | 0.589  | 0.98        | 0.985        |
| BBB    | volume + submission           | 0.847        | 0.591  | 0.98        | <b>0.990</b> |
| BBB    | journey + volume + submission | <b>0.855</b> | 0.574  | 0.98        | 0.985        |
| GGG    | volume                        | 0.604        | 0.216  | 0.61        | 0.540        |
| GGG    | journey + volume              | 0.549        | 0.643  | 0.65        | 0.605        |
| GGG    | <b>submission</b>             | 0.826        | 0.392  | 0.85        | 0.640        |
| GGG    | volume + submission           | 0.781        | 0.392  | 0.84        | 0.640        |
| GGG    | journey + volume + submission | <b>0.832</b> | 0.369  | <b>0.92</b> | 0.650        |

**The largest single improvement measured in this repository:** added to the previous recommendation, submission moves precision +0.092 on BBB and +0.283 on GGG, and p@100 by +0.06 and +0.27.

**Four scalars get almost all of it.** `submission` alone reaches 0.831 against the best arm's 0.855 on BBB and 0.826 against 0.832 on GGG — `submissionRate`, `missedAll`, `missedFirst`, `meanLateness` recover 97–99% of the full stack with no graph algorithm at all.

The `journey` figures here (0.549 GGG, 0.764 BBB) come from this harness, not from the GDS pipeline that produced the committed 0.722 and 0.838. **Cross-harness comparison against those numbers is not valid;** the five arms are comparable to each other because they share a harness, a split and a seed.

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## 8. Step 8 — persist the model, score a module it has never seen

`aga_fastpath_journeys.ipynb` step 15 trains and stores; `aga_score_unseen_module.ipynb` loads and scores.

### 8.1 What was persisted

```
RECOMMENDED_CUTOFF = 90
MODEL_NAME = f'oulad-atrisk-d{RECOMMENDED_CUTOFF}'
RECOMMENDED_FEATURES = [EMBEDDING_PROPERTY, 'logClicks',
                        'submissionRate', 'missedAll', 'missedFirst', 'meanLateness']
```

Trained on **BBB** and persisted with `gds.model.store()`, because **a session model dies with its session** (2-hour TTL). Before storing, the notebook checks the model flags a non-zero number of students — a GDS pipeline can return a majority-class classifier and report it as a successful train. It also refuses to store from a module with fewer than three assessments due by the cutoff, where four of the six features collapse to one boolean.

Two API facts this step turns on, both of which broke it first:

- **`gds.model.get(name)` returns `ModelDetails` — metadata.** Only the object returned by `train()` carries `.predict_stream` and `.drop`. A recovered model is applied by *name* through `gds.pipeline.node_classification.predict.stream(G, model_name, target_node_labels=[...])`.
- **drop and delete target different copies.** `gds.model.drop(name)` releases the session catalog entry; `gds.model.delete(name)` removes what `store()` persisted. Both take a name string, so neither accepts the `ModelDetails` that `get()` returns.

## 8.2 Scored on EEE — 2,632 students, 983 at risk, base rate 0.3735

The transferred model against a control retrained on EEE itself, precision at a fixed contact budget (`cost_of_transfer` is `refit - transferred`, so negative means transferring won):

| k   | transferred | refit_on_target | cost_of_transfer |
|-----|-------------|-----------------|------------------|
| 50  | 1.000       | 0.980           | -0.02            |
| 100 | 1.000       | 0.990           | -0.01            |
| 200 | 1.000       | 0.990           | -0.01            |
| 400 | 0.995       | 0.985           | -0.01            |

**Transfer is free** — transferring slightly beat retraining locally, which is within noise but rules out a transfer penalty. Retraining per module is not required.

The precision column, however, is an artefact and not the model's. `missedAll` alone flags 388 EEE students of whom 98.7% fail, so any ranker respecting that boolean is perfect through  $k = 200$ . A precision of exactly 1.000 across three budgets with an independently-retrained control landing within a hundredth of it is not two good models agreeing — it is two models reading the same column. That agreement was the tell.

## 8.3 The informative test — the 2,244 students `missedAll` does not flag

Base rate 0.2674:

| k   | precision | recall | lift |
|-----|-----------|--------|------|
| 50  | 0.98      | 0.082  | 3.67 |
| 100 | 0.96      | 0.160  | 3.59 |
| 200 | 0.93      | 0.310  | 3.48 |
| 400 | 0.69      | 0.460  | 2.58 |

A hundred-student worklist that is 96% correct where a quarter of students fail, produced by a model trained on a different module. Precision collapses past  $k \approx 300$ , so the usable budget is about 200.

Two conditions make this work and both are easy to lose: the activity-type vocabulary must be enumerated **globally** (§4.1), and the submission features must stay **scale-free** — a ratio, two booleans and a day count — so they carry across modules with different assessment counts.

## 8.4 The ablation on EEE, which isolates the embedding

The Phase 3 arms were re-run on EEE, both locally (train on EEE, test on an EEE holdout) and transferred (train on all of BBB, test on all of EEE), with both modules re-exported on the shared global vocabulary. Transfer, hard population (n = 2,244, base rate 0.2674):

| features                      | k=50        | k=100       | k=200        |
|-------------------------------|-------------|-------------|--------------|
| volume                        | 0.88        | 0.73        | 0.640        |
| journey + volume              | 0.68        | 0.62        | 0.535        |
| submission                    | 0.94        | 0.97        | 0.935        |
| <b>volume + submission</b>    | <b>1.00</b> | <b>0.98</b> | <b>0.945</b> |
| journey + volume + submission | 0.96        | 0.94        | 0.940        |

**On EEE the embedding does not contribute — it costs.** Against `volume + submission` it is  $-0.040 / -0.040 / -0.005$  transferred and  $-0.020 / -0.010 / -0.005$  locally, and the pattern repeats in all four cells (local and transfer  $\times$  full and hard). More pointedly, `journey + volume` at 0.68 is *worse than* `volume alone` at 0.88: adding 128 dimensions to a single scalar degrades it.

So the lift of 3.6 is real and it belongs to `volume + submission`. This also refutes the explanation offered for the GGG/BBB split — that the embedding earns more where assessment evidence is thin. EEE has two assessments by the cutoff, which is thin by that logic, and the embedding still loses. Whatever distinguishes GGG, assessment density is not it.

**What still holds.** Cross-module transfer works: `volume + submission` trained on BBB scores EEE at 1.00 / 0.98 / 0.945 on the hard population, so a single stored model does serve an unseen module. The finding is about which features earn their place, not about whether transfer is possible. One module, one seed, EEE only.

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## 9. Where the recommendation stands

**The stored model.** `oulad-atrisk-d90` — a GDS node classification pipeline over `journeyEmbedding + logClicks + submissionRate + missedAll + missedFirst + meanLateness`, cut at day 90, trained on BBB, logistic regression and random forest as candidates selected on F1 macro over four validation folds.

**On held-out data:** precision 0.855 on BBB and 0.832 on GGG — the top-precision arm in both modules — and p@100 of 0.92 on GGG. Under transfer to EEE it produces a  $\sim 200$ -student worklist at roughly  $3.5\times$  lift on the population where the cheap boolean is uninformative.

**Day 90, not day 30.** An earlier revision recommended a day-30 worklist on 0.923 precision. Held out that is 0.482 on GGG — a coin flip on the flagged set — against 0.702 on BBB. Day 90 is the cutoff that worked in both.

**One open tension, and it is the newest evidence.** On EEE the journey embedding subtracts from that six-feature set: `volume + submission` reaches 1.00 / 0.98 / 0.945 where the full stack gets 0.96 / 0.94 / 0.940. The stored model still carries the embedding. On current evidence a five-feature version without it is at least as good on EEE and marginally worse on BBB ( $-0.008$  precision), so the embedding's place rests on GGG alone

(+0.051) and the assessment-density explanation for that has been refuted. **Resolving this is the next measurement, not a conclusion this document can draw.**

## 9.1 What ships first, and it is not this model

**The zero-activity rule.** Precision 0.736 at day 7 against a 0.473 base rate, no graph analytics, 83 days of lead time. Then the submission trigger: precision 0.74–0.93 firing 15–59 days before the model. Both are fixed rules with nothing to fit, both reconcile against the CSVs, and the model performs *better* on the population left after the rule takes its share.

The order is cheapest-and-earliest first, with the model as the last and most expensive layer over students who start and then fade.

## 9.2 Not recommended

**The whole-journey model**, despite 0.902 holdout accuracy and 0.971 holdout precision. It needs the journey finished, and once a presentation ends `finalResult` is already in the data — it predicts something you know.

**FastRP**, worth +0.4 accuracy points over one logged aggregate.

**Volume alone, early.** It works retrospectively (0.825 and 0.766 on the whole journey) but is beaten in the window that matters: 0.382 and 0.417 at days 30 and 90 on GGG, 0.628 and 0.704 on BBB, against 0.482/0.722 and 0.702/0.838 for the embedding. It remains the honest floor to measure against — it is the one baseline here whose numbers never needed a holdout correction.

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## 10. Mistakes, and what they cost

The wrong turns are the most reusable part of this record.

**Measuring clustering quality instead of predictive performance.** Two FastPath runs were judged on Louvain modularity — 0.5470 and 0.5122 — and both read as failure. Modularity scores how cleanly a similarity graph partitions, not whether the partition predicts anything. The same embedding, scored as a classifier feature, reached 0.9280. *Cost: a nearly-abandoned method and a written conclusion that had to be retracted.*

**Never passing the feature to the algorithm.** `sumClick` was copied onto the event nodes and projected into the session, then never handed to FastPath because `event_node_feature_vector_property` was unset. *Cost: an entire full-scale run measuring the wrong configuration.*

**Two wrong diagnoses of the same zero.** Weighted degree centrality returned 0.0 for every material. Blamed first on the projection dropping a property (disproved by probing a live session), then on `orientation='REVERSE'` needing an inverse index (disproved by adding one). The actual cause: `node_labels=['EducationalMaterial']` induces a subgraph of material nodes only, which discards every REVIEWED\_MATERIAL relationship, since each one's source is a Student. Degree over a relationship-free subgraph is zero, silently. *Cost: two sessions and a committed explanation that was wrong.*

**A rewritten query that changed its own grouping.** Dropping `s.id` from a baseline query's RETURN made `sum()` aggregate by the class label, returning 2 rows instead of 2,359. *Cost: one misleading comparison table, caught because a 0.5000 looked wrong.*

**Recommending on accuracy.** The first version called the day-60 model "modest, ~0.70 accuracy" and treated that as its ceiling. Measuring precision showed a much better worklist — the model is conservative and accuracy penalises exactly that. *Cost: a recommendation pitched below what the model can do.*

**Then recommending on in-sample precision.** The replacement precision was measured with `predict_stream` over every labelled student, including the 70% trained on. Held out properly, day 90 fell from 0.857 to 0.722 and day 30 from 0.923 to 0.482. Worse, the inflation was dismissed by pointing at a small accuracy gap, when accuracy is the metric least sensitive to memorisation on an imbalanced target. *Cost: a recommendation that would have put a coin-flip model in front of an intervention team.*

**Citing a hindsight statistic as decision-point evidence.** The 88% withdrawal figure for never-engaged students, 61.4% of whom had already quit before day 0. *Cost: a day-14 rule proposed on the strength of a number that could not support it.*

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## 11. What caught them

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Worth keeping in any similar exercise:

- **A trivial baseline in the notebook, not in your head.** Every notebook trains or computes one. Most of the wrong conclusions above were caught by a single aggregate doing better.
  - **Ablations built in.** Training embedding-only, feature-only and both is three runs of a few seconds and is the only way to attribute a gain.
  - **A holdout you built yourself.** Every inflated figure in this record came from evaluating over a population that included training rows.
  - **Cross-check the session against the database.** Comparing a session's weighted degree against `sum(r.sumClick)` from AuraDB is what surfaced the `node_labels` bug — the gap was 420,908 before the fix and 0 after.
  - **Guards that abort rather than warn.** A missing relationship weight is not an error in GDS, it is a silent zero. Projections print what they actually carry and stop if a needed property is absent.
  - **Anchor points in a sweep.** Including a cutoff whose answer is already known turns a shortcut into a self-validating one.
  - **Reconcile a graph query against the source CSVs.** 176 of 176 exact for submission; four students out for the zero-activity rule, with the discrepancy explained.
- 

## 12. Before any of this ships

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**Validate on the remaining four modules.** GGG, BBB and EEE are measured. AAA, CCC, DDD and FFF are not, and the three measured modules disagree by 12 precision points and by whether the embedding helps at all.

**Resolve the embedding's place.** Re-run the five-arm ablation on GGG and BBB against the same global vocabulary EEE used, and decide whether `oulad-atrisk-d90` should keep `journeyEmbedding`.

**Finer cutoffs.** [20, 30, 40, 50, 60] would locate the onset of signal more precisely than the current 30-to-60 bracket. Cheap, and more informative than another module.

**Nothing here is temporal.** Presentations are pooled rather than held out in time order, so no result speaks to whether a model trained on 2013 presentations predicts 2014 ones.

**Precision is not actionability.** A student who has missed the first assessment may already be past recovery. Nothing measured here establishes that contacting these students changes anything.

**A flag is not a diagnosis.** The rules cannot distinguish a student who has dropped out in all but name from one who registered late or is studying offline.

### 13. Reproducing

| artefact                               | what it does                                              | knobs                                                      |
|----------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|
| oulad_data_load.ipynb                  | loads the graph (66,920 nodes, 8,818,076 relationships)   | —                                                          |
| aga_outcome_prediction.ipynb           | FastRP + classification, with ablation                    | MODULE, PASS_RESULTS, VARIANTS                             |
| aga_fastpath_journeys.ipynb            | FastPath + classification, cutoff sweep, stores the model | MODULE, CUTOFF_DAY, RUN_SWEEP, SWEEP_CUTOFFS, DELETE_CHAIN |
| aga_score_unseen_module.ipynb          | loads the stored model, scores an unseen module           | TARGET_MODULE, CONTACT_BUDGET, CUTOFF_DAY                  |
| scripts/zero_activity_rule.py          | the day- <i>D</i> rule, offline                           | thresholds in-file                                         |
| scripts/assessment_submission.py       | submission trigger, phase 1, offline                      | —                                                          |
| scripts/assessment_submission_graph.py | phase 2 reconciliation, exits 1 on gate failure           | —                                                          |
| scripts/assessment_submission_model.py | phase 3, five arms                                        | —                                                          |

All analytics notebooks delete their sessions, and the FastPath one deletes its event chain and the two `Student` properties it writes. The graph returns to 66,920 nodes and 8,818,076 relationships, verified after every run in this record.

The two notebooks that hold a driver open across long algorithm runs pass `liveness_check_timeout=30`, `max_connection_lifetime=600`. Without it a stale pooled connection surfaces as *"Unable to retrieve routing information"* in whichever cell runs next — on one run that cell was the cleanup step, which left 597,336 orphan `Interaction` nodes in the database.